ReDI School of Digital Integration

ExoDetect:

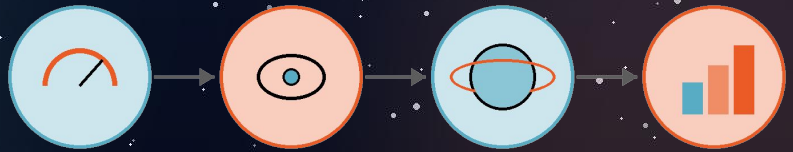
From Tiny Light Dips to Possible New Worlds

Can machine learning help us decide which Kepler candidates deserve a second look?

We use tech to connect human potential and opportunity with dignity & humility

What if one tiny shadow is a new world?!

- Kepler was a space telescope that searched for exoplanets.
- It did this by watching the brightness of stars.
- When a planet passes in front of its star, the light becomes a little bit weaker.
- This tiny drop in brightness can be a clue that a planet exists.
- Not every light dip is a real planet; some signals are false positives.
- Machine learning can help rank the most promising candidates for further study.



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Kepler Dataset & ML Setup

What data was used, what features were selected, and what EDA was done?

1 Dataset Snapshot

Total KOIs	Confirmed	False Positives	Candidates	Labeled training pool used for classification
9,564	2,747	4,839	1,978	7,586

2 ML Goal in This Project

- Train a classifier on known labels: CONFIRMED vs FALSE POSITIVE
- Predict planet-like probabilities for unresolved CANDIDATES
- Rank candidates for follow-up priority

3 Selected Features

Feature	Meaning
koi_period	orbital period
koi_prad	planet radius
koi_duration	transit duration
koi_depth	transit depth
koi_model_snr	signal-to-noise ratio
koi_impact	transit geometry / impact parameter
koi_teq	equilibrium temperature
koi_insol	incident stellar flux
koi_steff	stellar effective temperature
koi_slogg	stellar surface gravity
koi_srad	stellar radius
koi_kepmag	Kepler magnitude

4 EDA Performed

- Class distribution of KOI labels
- Missing values in selected features
- Radius vs orbital period
- Planet radius distribution by disposition
- Signal-to-noise ratio vs radius
- Correlation heatmap of selected numerical features

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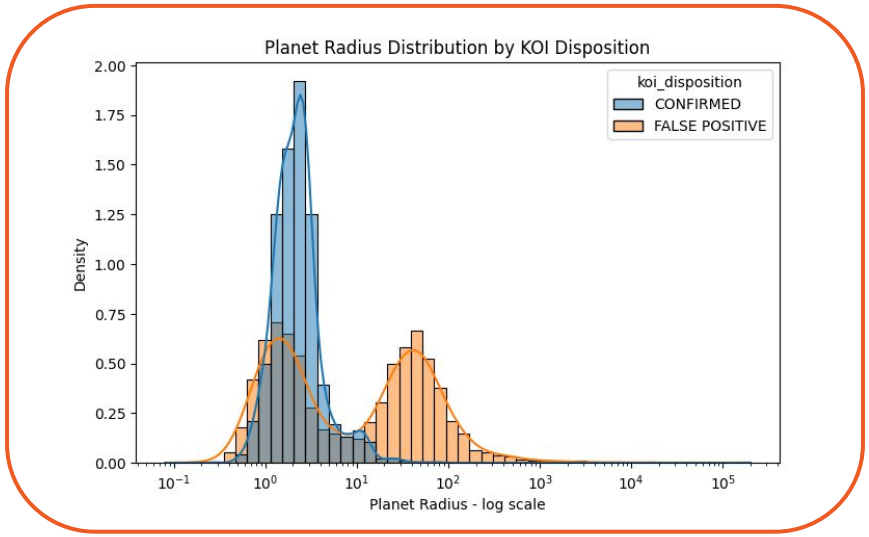
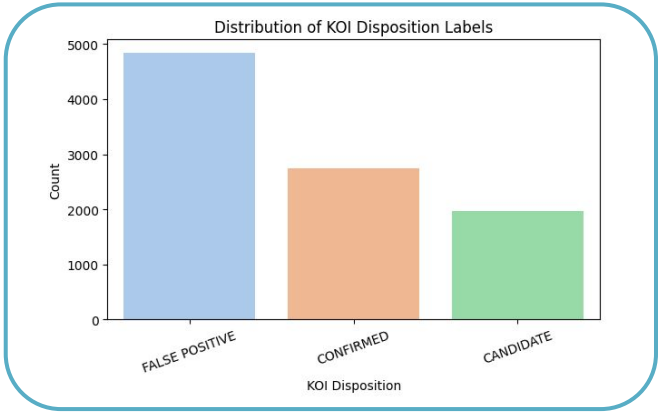
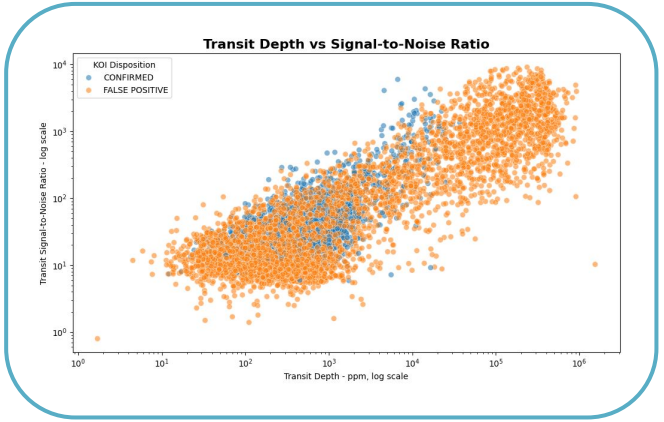
Can the data give us scientific clues before ML ?







- Radius is a useful clue: false positives often show larger or more extreme radius values.
- The dataset is imbalanced, with more false positives than confirmed planets or candidates.
- Transit depth and signal strength show patterns, but the classes still mix, so ML is needed.

Which ML model performed best?

Classification task: CONFIRMED vs FALSE POSITIVE

Training pipeline

What happened before candidate ranking?

- **Selected features** — radius, period, depth, SNR...
- **Train / test split** — test data was unseen
- **Fit several models** — LR, SVM, GB, RF
- **Compare scores** — Macro F1 + ROC-AUC
- **Select final model** — Tuned Random Forest

Model comparison on test set

Model	Accuracy	F1 Macro	ROC-AUC
Random Forest	**0.934**	**0.929**	**0.978**
Gradient Boosting	0.926	0.920	0.976
SVM	0.847	0.842	0.931
Logistic Regression	0.828	0.823	0.903

Selected model

Tuned Random Forest

Why these metrics?

Macro F1 handles imbalance
ROC-AUC supports probabilities

Best model selected for the next step: predicting planet-like probabilities for unresolved CANDIDATES.

How did classification become candidate ranking?

The ML model first learns a classification task, then its probabilities are used to rank unresolved Kepler candidates.

From training to ranking

Important distinction: CANDIDATES were not used to train the classifier.

1 Train on known labels — ML training
CONFIRMED = 1 vs FALSE POSITIVE = 0

2 Check on unseen test data — Evaluation
The test set has real labels, so scores can be measured.

3 Apply to unresolved CANDIDATES — Prediction use
These objects have no confirmed answer in the dataset.

4 Create planet_probability — ML output
Probability of class 1: P(CONFIRMED | selected features).

5 Sort candidates by probability — Ranking
Highest probability = most planet-like shortlist item.

Exact step in the notebook

```
X_candidates = candidate_df[feature_cols]
planet_probability = best_pipeline.predict_proba(X_candidates)[:, 1]
rank = sort by planet_probability ↓
```

[:, 1] means the probability of the positive class: CONFIRMED.

Top candidate ranking preview

Candidate	Planet-like probability	Radius	SNR
K01145.01	0.993	2.32	33.1
K00426.01	0.987	3.84	62.4
K01972.01	0.987	2.88	39.8
K01861.01	0.987	3.56	47.5
K04526.01	0.983	2.72	15.0

Final output: Not official planet confirmation , a smart shortlist for follow-up study.

Which unresolved candidates look most planet-like?

The trained model ranks unresolved Kepler candidates by predicted planet-like probability.

Why is it interesting?

1. High model probability
2. Reasonable planet radius
3. Strong transit signal / Good follow-up target

Top candidate ranking preview

Candidate	Probability	Radius	SNR	Priority
KOI-K01145.01	99.3	2.32	33.1	High Priority
KOI-K00426.01	98.7	3.84	62.4	High Priority
KOI-K01972.01	98.7	2.88	39.8	High Priority
KOI-K01861.01	98.7	3.56	47.5	High Priority
KOI-K04526.01	98.3	2.72	15.0	High Priority

Probability = model confidence that a candidate is planet-like (class 1: CONFIRMED).

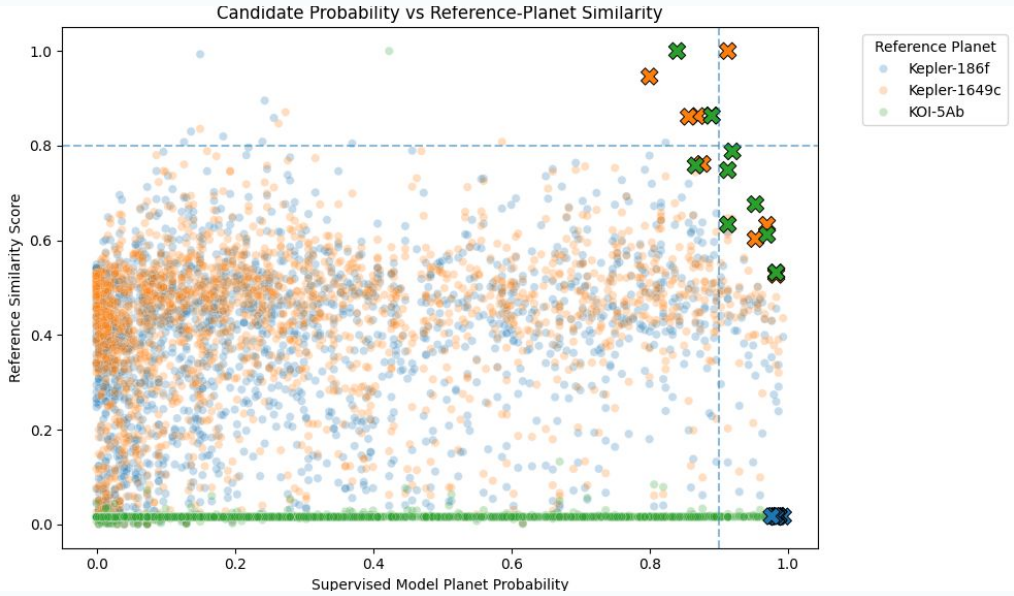
Top candidate spotlight

- Candidate: KOI-K01145.01
- ✓ Planet-like probability: 99.3
- Radius: 2.32

Final output: A prioritized shortlist for follow-up observation.

Reference similarity: a smarter shortlist

High model probability + high similarity to known planets = stronger follow-up targets.



What this adds

Score = 70% probability + 30% similarity

1. Model probability stays the main signal.
2. Similarity highlights candidates close to known planets.
3. Upper-right points become follow-up targets, not official discoveries.

Example: K02159.01 ranked high and is still a candidate → useful for future follow-up.

Takeaway: ExoDetect does not confirm planets — it prioritizes promising KOIs for deeper checks with light curves, TESS data, and expert follow-up.

Can AI discover planets?

Not alone — but it can help us decide where to look first.

✓ What worked

- Built a full ML pipeline
- Compared several models
- Ranked unresolved candidates

! Limitations

- Not official planet confirmation
- No raw light curve analysis
- Based on tabular Kepler features

→ Future work

- Use raw light curves
- Validate with TESS data
- Add explainability with SHAP

Final answer: **ML turns thousands of signals into a smarter shortlist for follow-up study.**

Thank You!

Questions?

Thank you for your time, support, and attention.

Let's connect on LinkedIn — scan the QR code.

Dalileh Oladzadeh

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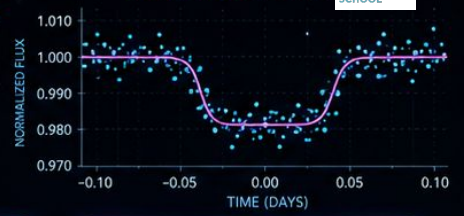
Scan to connect on LinkedIn

Evaluation Metrics Summary

For Exoplanet Classification Models

KEPLER DATA
CANDIDATE TARGET
KIC 8462852



1 Accuracy

Overall proportion of correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2 Precision

Of predicted positives, how many are truly positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

3 Recall

Of actual positives, how many did the model find?

$$\text{Recall} = \frac{TP}{TP + FN}$$

4 F1 Score

Balanced measure of Precision and Recall

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$$

5 AUC-ROC

Measures separability across classification thresholds

ROC: plot of TPR vs FPR

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

AUC = Area Under the ROC Curve

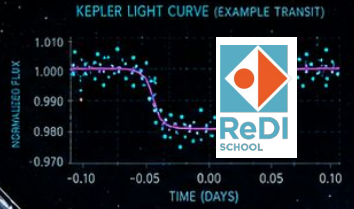
TP = True Positive | **TN** = True Negative | **FP** = False Positive | **FN** = False Negative

KEPLER DATA
CANDIDATE TARGET
KIC 8462852

Models Used in My Project

How each model works + a key formula

Kepler Exoplanet Candidate Classification



1 Decision Tree

Splits the data step by step using the best feature threshold.

$$\text{Information Gain} = H(\text{parent}) - \sum (n_j / n)\, H(\text{child}_j)$$

★ Chooses the split with highest information gain.

2 Random Forest

Builds many decision trees and combines their votes.

$$\hat{y} = \text{majority\_vote}(T_1(x), T_2(x), \ldots, T_B(x))$$

★ Reduces overfitting and improves stability.

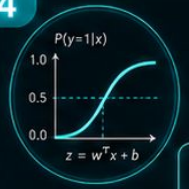
3 XGBoost

Adds trees sequentially to correct previous errors.

$$\hat{y}_i = \sum f_t(x_i)$$

★ Boosting model with strong performance on tabular data.

4 Logistic Regression



Estimates the probability of the positive class.

$$P(y=1|x) = 1 / (1 + e^{-(w^T x + b)})$$

★ Uses the sigmoid function for classification.

5 KNN

Looks at the nearest samples and predicts by neighbor vote.

$$\hat{y} = \text{mode}(\{y_i : x_i \in N_k(x)\})$$

★ Prediction depends on the k nearest neighbors.

6 SVM

Finds the boundary that maximizes the margin between classes.

$$f(x) = \text{sign}(w^T x + b)$$

★ Aims for the widest possible separating margin.

ML FOR EXOPLANET DETECTION

x = input features | $\hat{y}$ = predicted class | H = entropy | w, b = model parameters